

Advanced Statistical Mechanics Mid-Term Cheat-Sheet Formulas

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- Thermal equilibrium

$$\Theta(A) = \Theta(B)$$

- Equation of state (example)

$$\Theta(P, V) = \text{const}$$

- First Law (finite)

$$\Delta E = Q + W$$

- First Law (differential)

$$dE = \delta Q + \delta W$$

- Adiabatic process

$$\delta Q = 0$$

- Isothermal process

$$dT = 0$$

- Cyclic process

$$\oint dE = 0$$

- Exact vs inexact

$$dE \text{ exact, } \delta Q, \delta W \text{ inexact}$$

- Generalized work

$$\delta W = \sum_i J_i dx_i$$

- Pressure–volume work

$$\delta W = -P dV$$

- Magnetic work

$$\delta W = H dM$$

- Electrical work

$$\delta W = E dP$$

- Chemical work

$$\delta W = \mu dN$$

- Ideal gas law

$$PV = nRT$$

- Internal energy (ideal gas)

$$E = E(T)$$

- Monatomic ideal gas

$$E = \frac{3}{2} N k_B T$$

- General ideal gas

$$E = \frac{f}{2} N k_B T$$

- Joule free expansion

$$\Delta E = 0$$

- Isothermal ideal gas

$$\Delta E = 0 \Rightarrow Q = W$$

- Heat capacity

$$C = \frac{\delta Q}{dT}$$

- Constant volume

$$C_V = \left(\frac{\partial E}{\partial T} \right)_V$$

- Constant pressure

$$C_P = \left(\frac{\partial E}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P$$

- Ideal gas relation

$$C_P - C_V = nR$$

- Thermal expansion

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P$$

- Isothermal compressibility

$$\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T$$

- Adiabatic compressibility

$$\kappa_S = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_S$$

- Kelvin statement

$$\eta < 1$$

- Clausius statement

$$\delta Q_{C \rightarrow H} \neq 0 \text{ without work}$$

- Energy balance (cycle)

$$W = Q_H - Q_C$$

- Efficiency

$$\eta = \frac{W}{Q_H}$$

- Refrigerator coefficient

$$\omega = \frac{Q_C}{W}$$

- Carnot efficiency

$$\eta_C = 1 - \frac{T_C}{T_H}$$

- Carnot heat ratio

$$\frac{Q_C}{Q_H} = \frac{T_C}{T_H}$$

- Carnot work

$$W = Q_H \left(1 - \frac{T_C}{T_H} \right)$$

- Adiabatic condition

$$\delta Q = 0$$

- Adiabatic equation

$$PV^\gamma = \text{const}$$

- Exponent

$$\gamma = \frac{C_P}{C_V} = \frac{f+2}{f}$$

- Temperature–volume relation

$$TV^{\gamma-1} = \text{const}$$

- Temperature–pressure relation

$$T^\gamma P^{1-\gamma} = \text{const}$$

- Clausius inequality

$$\oint \frac{\delta Q}{T} \leq 0$$

- Reversible cycle

$$\oint \frac{\delta Q_{\text{rev}}}{T} = 0$$

- Entropy definition

$$dS = \frac{\delta Q_{\text{rev}}}{T}$$

- Entropy change

$$\Delta S = \int \frac{\delta Q_{\text{rev}}}{T}$$

- Irreversible process

$$\Delta S > \int \frac{\delta Q}{T}$$

- Entropy of isolated system

$$\Delta S \geq 0$$

- Fundamental thermodynamic identity

$$dE = T dS + \sum_i J_i dx_i$$

- Temperature definition

$$T = \left(\frac{\partial E}{\partial S} \right)_{x_i}$$

- Generalized forces

$$J_i = \left(\frac{\partial E}{\partial x_i} \right)_S$$

- Boltzmann entropy

$$S = k_B \ln \Omega$$